MTH2003 Tutorial Sheet 2 Solutions

1. **Question** $y'' - 6y' + 9y = 10\,\mathrm{e}^{3t}$.

   Solution

   [i] Constant coefficients case: the characteristic equation $m^2 - 6m + 9 = 0$ has a double root $m = 3$, so the complementary solution is $\boxed{y_c = (C_1 + C_2 t)\,\mathrm{e}^{3t}}$.

   [ii] The index in the exponential on the right-hand side is $\alpha = 3$ which coincides with *the double root* of the characteristic equation. Hence we have a *double* ($s = 2$) resonance and we seek a particular solution of the form $y_p = Bt^2\,\mathrm{e}^{3t}$. Substituting into the ODE gives:

   $$\text{LHS} = \left[9Bt^2\,\mathrm{e}^{3t} + 12Bt\,\mathrm{e}^{3t} + 2B\,\mathrm{e}^{3t}\right] - 6\left[3Bt^2\,\mathrm{e}^{3t} + 2Bt\,\mathrm{e}^{3t}\right] + 9Bt^2\,\mathrm{e}^{3t} = \text{RHS} = 10\,\mathrm{e}^{3t}.$$

   We can see that the $t^2\,\mathrm{e}^{3t}$ terms and the $t\,\mathrm{e}^{3t}$ terms cancel out. We are left with $2B = 10$, so $B = 5$, and the particular solution is $\boxed{y_p = 5t^2\,\mathrm{e}^{3t}}$.

   [iii] The General Solution is $\boxed{y = (C_1 + C_2 t + 5t^2)\,\mathrm{e}^{3t}}$.

2. **Question** *Prove that for any p and q*

$$\begin{aligned} \mathrm{e}^p + \mathrm{e}^q &= 2\mathrm{e}^{(p+q)/2}\cosh[(p-q)/2], \\ \mathrm{e}^p - \mathrm{e}^q &= 2\mathrm{e}^{(p+q)/2}\sinh[(p-q)/2]\,. \end{aligned}$$

Hence show that if a, b and c are constants that satisfy $b^2 > 4ac$ then any solution of the ODE

$$ay'' + by' + cy = 0$$

can be written in the form $y(t) = \mathrm{e}^{\alpha t}(A\cosh\beta t + B\sinh\beta t)$. Find α and β in terms of a, b and c.

Solution Note that

$$\begin{aligned} 2\mathrm{e}^{(p+q)/2}\cosh[(p-q)/2] &= \mathrm{e}^{(p+q)/2}(\mathrm{e}^{(p-q)/2} + \mathrm{e}^{(q-p)/2}) \\ &= \mathrm{e}^p + \mathrm{e}^q \end{aligned}$$

and similarly

$$\begin{aligned} 2\mathrm{e}^{(p+q)/2}\sinh[(p-q)/2] &= \mathrm{e}^{(p+q)/2}(\mathrm{e}^{(p-q)/2} - \mathrm{e}^{(q-p)/2}) \\ &= \mathrm{e}^p - \mathrm{e}^q. \end{aligned}$$

We find a solution of $ay'' + by' + cy = 0$ by considering $y = \mathrm{e}^{mt}$ and substituting to give

$$(am^2 + bm + c)\mathrm{e}^{mt} = 0$$

which means that, for all t we need $am^2 + bm + c = 0$ and so $m_\pm = -b/(2a) \pm \sqrt{b^2 - 4ac}/(2a)$. Hence for $b^2 > 4ac$ we have real roots and

$$y = C\mathrm{e}^{m_+ t} + D\mathrm{e}^{m_- t}$$

which can be written

$$y = \frac{C+D}{2}(\mathrm{e}^{m_+ t} + \mathrm{e}^{m_- t}) + \frac{C-D}{2}(\mathrm{e}^{m_+ t} - \mathrm{e}^{m_- t}).$$

Using the identity from the first part we have

$$y = (C+D)\mathrm{e}^{\alpha t}\cosh\beta t + (C-D)\mathrm{e}^{\alpha t}\sinh\beta t$$

where $\alpha = (m_+ + m_-)/2 = -b/(2a)$ and $\beta = (m_+ - m_-)/2 = \sqrt{b^2 - 4ac}/(2a)$.

Alternatively: We can directly write

$$y = C\mathrm{e}^{(\alpha+\beta)t} + D\mathrm{e}^{(\alpha-\beta)t}$$

where $\alpha = -b/(2a)$ and $\beta = \sqrt{b^2 - 4ac}/(2a)$. This can be written in the form

$$\begin{aligned} y &= \mathrm{e}^{\alpha t}(C\mathrm{e}^{\beta t} + D\mathrm{e}^{-\beta t}) \\ \Rightarrow y &= \mathrm{e}^{\alpha t}(A\cosh\beta t + B\sinh\beta t). \end{aligned}$$

where $A = C + D$ and $B = C - D$.

3. **Question** *Consider again the equation from the Examples Sheet:*

$$2t^2y'' + 3ty' - y = 0\,, \quad t > 0\,.$$

On the Examples Sheet you were given the solution $y_1 = 1/t$ and you found a second solution y_2.

(***a***) **Question** *Write down Abel's identity for this equation.*
Answer The normalised form of the equation is

$$y'' + \frac{3}{2t}y' - \frac{1}{2t^2}y = 0$$

hence $p = 3/2t$, and Abel's identity states

$$W(t) = A\exp\left(-\int p(t)\,\mathrm{d}t\right) = A\exp\left(-\int \frac{3}{2t}\,\mathrm{d}t\right) = A\exp\left(-\frac{3}{2}\ln|t| + B\right) = Ct^{-3/2}, \qquad C = A\,\mathrm{e}^B, \quad t > 0$$

Question *Verify that y_1 and the y_2 that you found satisfy this identity.*
Answer For $y_1 = t^{-1}$, $y_2 = t^{1/2}$, we have

$$W(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t^{-1} & t^{1/2} \\ -t^{-2} & \frac{1}{2}t^{-1/2} \end{vmatrix} = \frac{3}{2}t^{-3/2}$$

in accordance with Abel's identity for $C = 3/2$.

(***b***) **Question** *Now imagine that you had not found y_2, and use the knowledge of y_1 and Abel's identity to find the general solution of this equation (the answer should be equivalent to that you found before, of course.)*
Answer Abel's identity written for the known solution y_1 and any solution y states

$$W[y_1, y] = \begin{vmatrix} y_1 & y \\ y_1' & y' \end{vmatrix} = \begin{vmatrix} t^{-1} & y \\ -t^{-2} & y' \end{vmatrix} = t^{-1}y' + yt^{-2} = Ct^{-3/2}.$$

This is a linear first-order equation and can be solved by the integrating factor method:

$$y' + t^{-1}y = Ct^{-1/2}, \quad t > 0$$

and so the integrating factor is

$$h = \exp\int\left(\frac{1}{t}\right)\,\mathrm{d}t = \exp\left(\ln t\right) = t$$

(as usual we do not need an integration constant here) and on multiplying the ODE by h we obtain

$$ty' + y = Ct^{1/2}$$

$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left(ty\right) = Ct^{1/2}$$

$$\Rightarrow ty = C\int t^{1/2}\mathrm{d}t = \frac{2C}{3}t^{3/2} + D.$$

and finally

$$y = Et^{1/2} + Dt^{-1}$$

as before (up to the names of the arbitrary constants).

4. **Question** *On the Examples Sheet you found the general solution of the following nonhomogeneous, second order, linear ODEs:*

$$y'' + 6y' + 8y = 16e^{-2t}$$

and

$$y'' + 6y' + 8y = e^{3t}\sin(2t)\,.$$

Using those results, write down the general solution of the ODE

$$y'' + 6y' + 8y = e^{-2t} + 1537e^{3t}\sin(2t)\,.$$

Solution The linear operator on the left-hand side is the same as in questions 1b and 1e from the Examples Sheet, so the complementary solution is the same as there:

$$y_c = C_1\,\mathrm{e}^{-2t} + C_2\,\mathrm{e}^{-4t}.$$

The right-hand side here is a linear combination of those in parts 1b and 1e. Namely, if we denote the right-hand side of 1b as $R_b = 16\,\mathrm{e}^{-2t}$, and the right-hand side of 1e as $R_e = \mathrm{e}^{3t}\sin(2t)$, then present right-hand side $R = \mathrm{e}^{-2t} + 1537\mathrm{e}^{3t}\sin(2t)$ is a linear combination of the previous two,

$$R = \frac{1}{16}R_b + 1537R_e.$$

Hence, by the superposition principle for nonhomogeneous equations (Theorem 3.20), the present particular solution can be taken as a similar linear combination of the previous particular solutions. That is, if we denote the particular solution from part 1b as $y_b = 8t\,\mathrm{e}^{-2t}$ and the particular solution from part 1e as $y_e = \frac{1}{1537}\,\mathrm{e}^{3t}\,(31\sin(2t) - 24\cos(2t))$, the particular solution y_p for the present example can be taken as

$$y_p = \frac{1}{16}y_b + 1537y_e = \boxed{\frac{1}{2}t\,\mathrm{e}^{-2t} + \mathrm{e}^{3t}\,(31\sin(2t) - 24\cos(2t))\,.}$$

The general solution is therefore:

$$\boxed{y = \frac{1}{2}t\,\mathrm{e}^{-2t} + \mathrm{e}^{3t}\,(31\sin(2t) - 24\cos(2t)) + C_1\,\mathrm{e}^{-2t} + C_2\,\mathrm{e}^{-4t}.}$$